
Virtual Private Networks (VPNs)

Virtual Private Network: Definition

- ▷ A VPN extends a security perimeter
 - ◆ Without reducing its overall security
 - ◆ In many cases it allows something similar to (virtual) link security through an insecure WAN (Internet)
- ▷ Examples
 - ◆ Remote access to an organizational network (e.g. UA)
 - ◆ Interconnection of two organizational networks
- ▷ A VPN enables a user to access resources that are otherwise protected
 - ◆ Networks, services, devices, etc.
- ▷ Consequently, managing a VPN requires
 - ◆ Authentication of the users (VPN creators)
 - ◆ Secure communications
 - If involving insecure networks
 - ◆ Access control to services (VPN services, other services)
 - Authorization

VPN: Mechanisms (1/3)

▷ Secure tunneling

- ◆ Allows to encapsulate protocols within other protocols
- ◆ Protects every encapsulated protocol
- ◆ Allows the tunneling tool to perform traffic access control



VPN: Mechanisms (2/3)

▷ Mediation (proxying)

- ◆ Transparent exploitation of VPN links by intermediate hosts
- ◆ Routers / gateways are typical examples

▷ Firewalls

- ◆ They are a natural frontier of security perimeters
- ◆ A proper extension of a security perimeter often results from the combination of a firewall and a VPN management system

VPN: mechanisms (3/3)

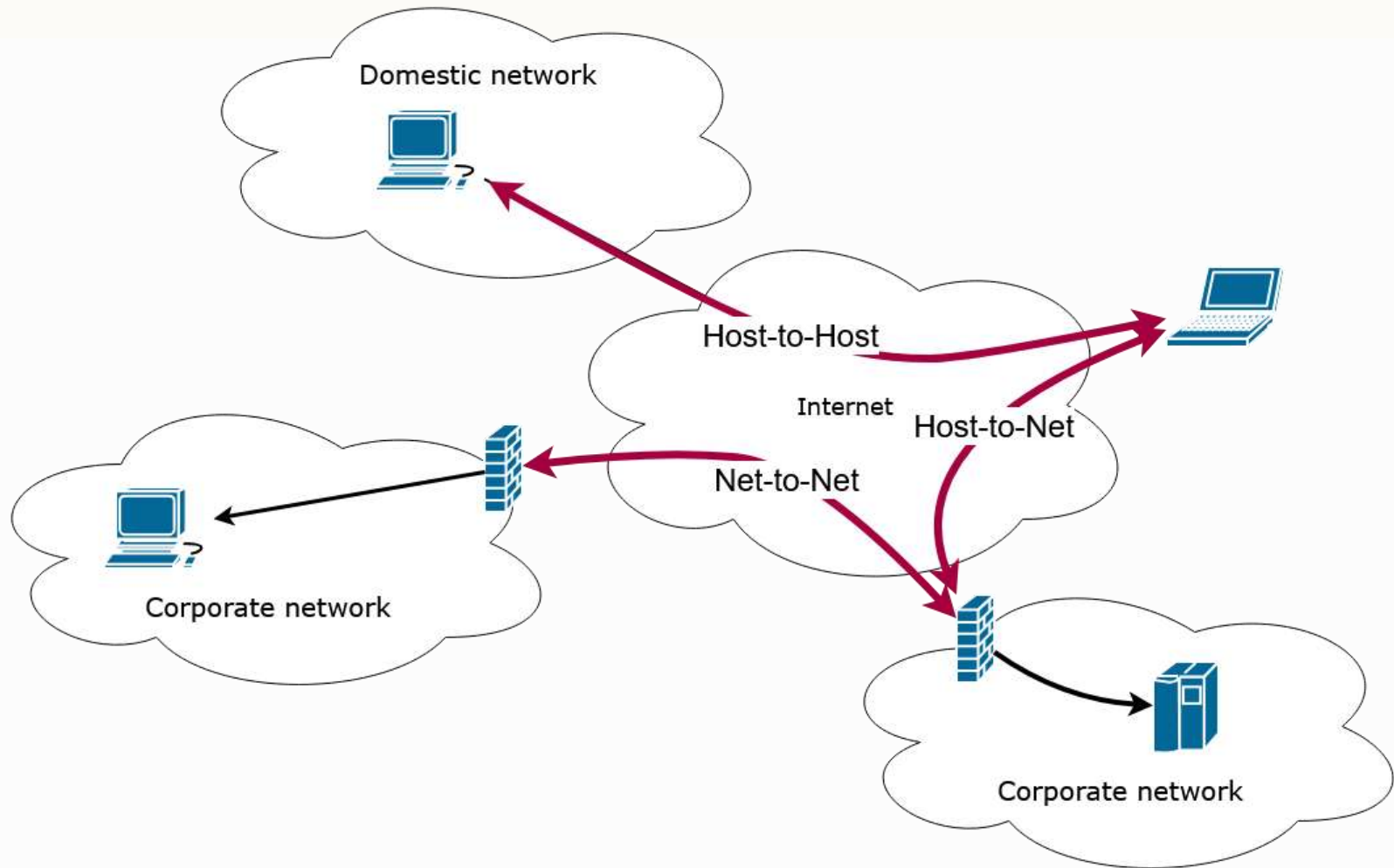
▷ User & network authentication

- ♦ Networks often have some authentication mechanism
 - Kerberos, RADIUS, 802.1X, etc.
- ♦ The VPN authentication protocol can use them or other ones

▷ VPN examples

Application	SSH SSL / TLS Tunnel	
Presentation		
Session		
Transport		
Network	IPSec, WireGuard	OpenVPN
Link	PPTP, L2TP	
Physical		

VPN types

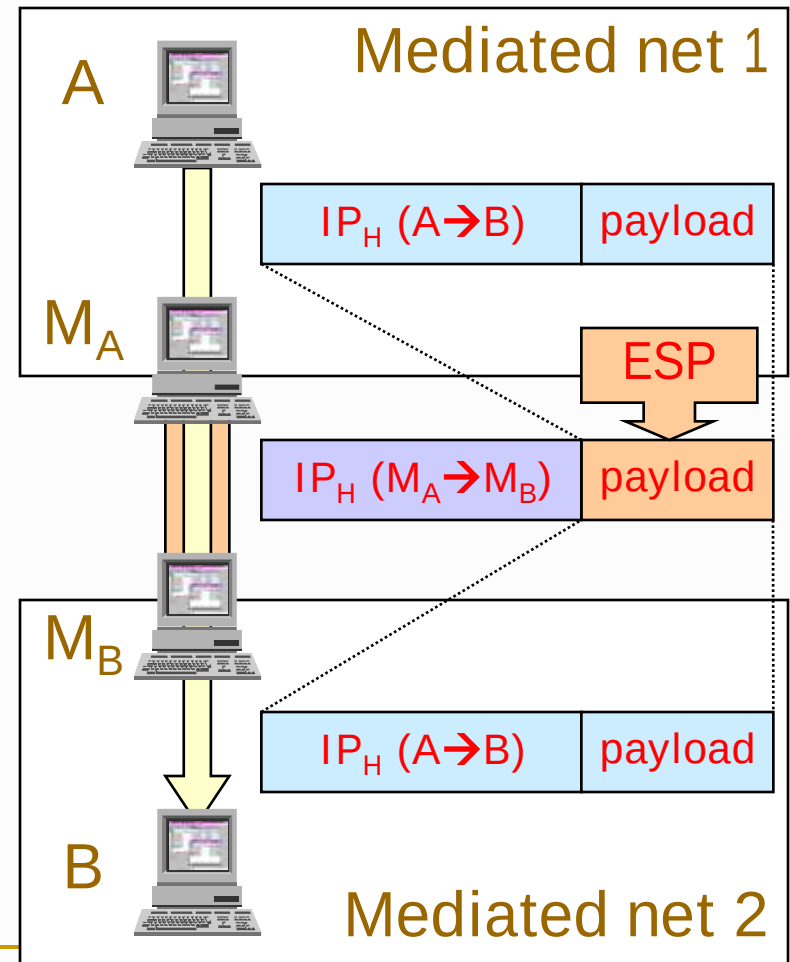


Mediation (proxying)

- ▷ Data circulates without protection in internal networks
 - ◆ Where the risk is controlled
 - ◆ This concept is being abandoned in Zero Trust Architectures!
- ▷ When data circulates through an insecure networks it is transformed to enforce its security
 - ◆ Confidentiality (header/payload encryption)
 - ◆ Integrity control (a MAC is added to check authenticity)
- ▷ Data is processed and validated
 - ◆ When pass through a mediator (proxy) controlling the access to/from a secure network
- ▷ Advantages
 - ◆ Mediated nodes don't need to manage VPNs
 - ◆ Traffic inspection has access to decrypted traffic in protected networks

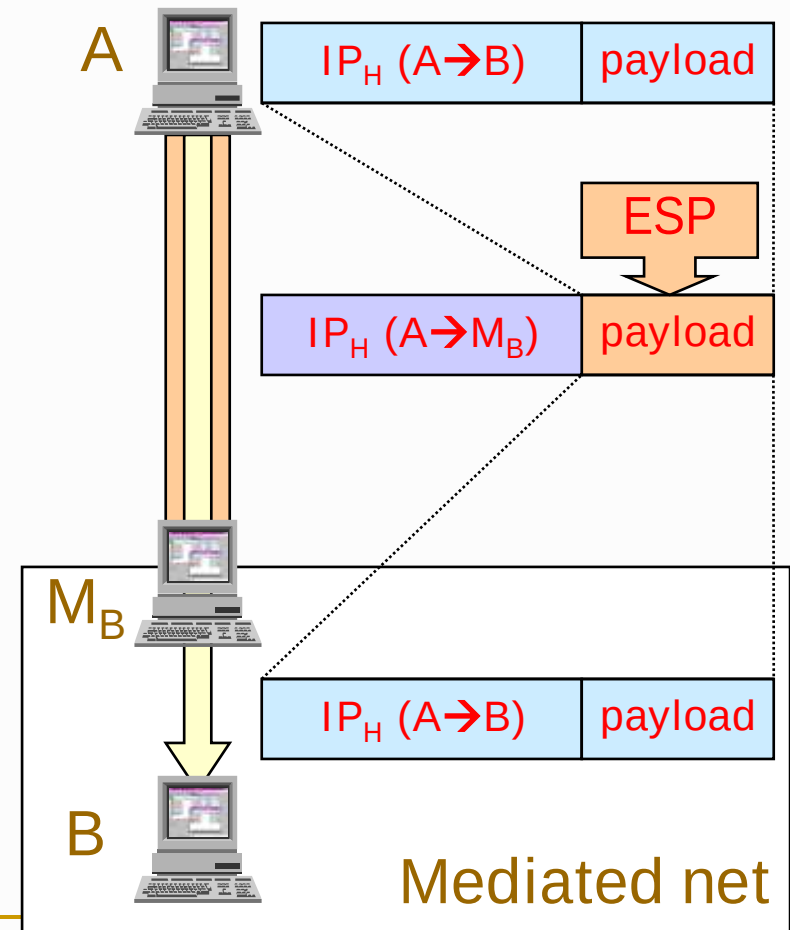
Net-to-net IPSec VPN: Mediation with ESP tunnel mode

- ▷ IPSec between M_A and M_B
 - ♦ A and B don't have to use IPSec
 - ♦ A and B addresses may be private
- ▷ M_A and M_B take the initiative to set up and use an IPSec tunnel
 - ♦ This is transparent to A and B

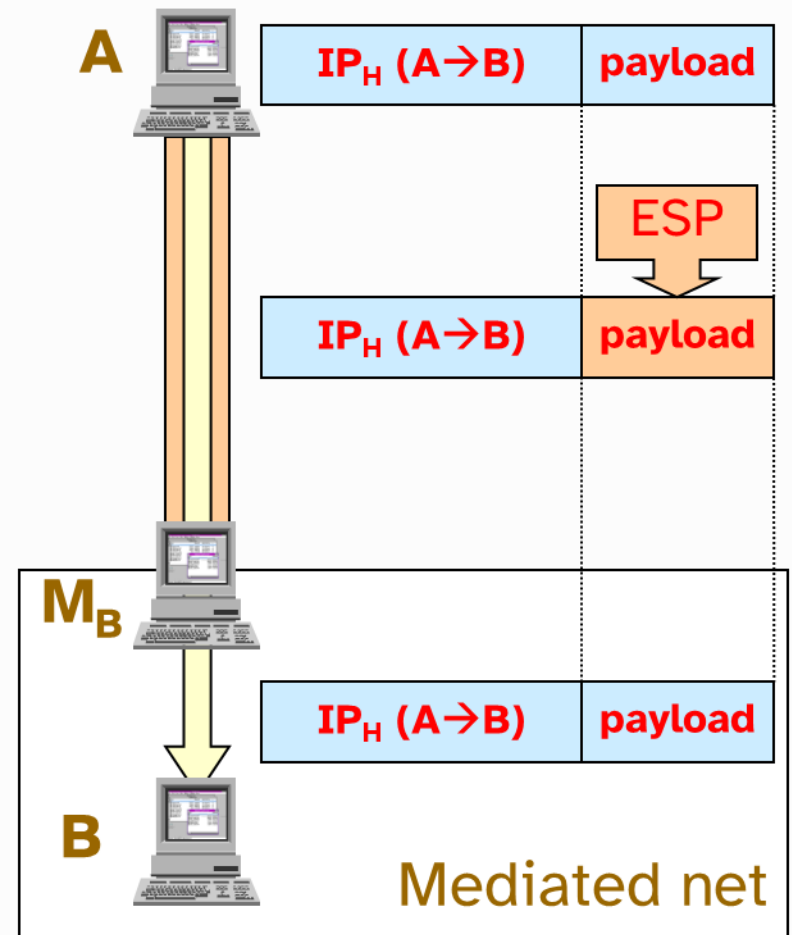
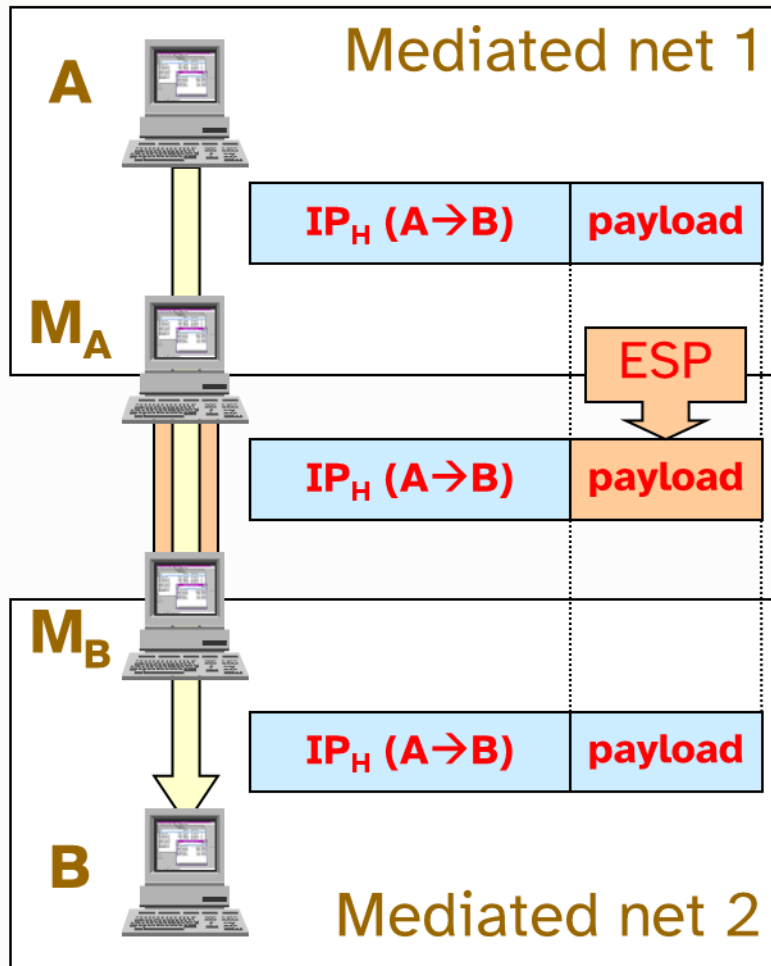


Host-to-net IPSec VPN: Mediation with ESP tunnel mode

- ▶ IPSec between A and MB
 - ♦ B doesn't need to use IPSec
 - ♦ B address may be private
- ▶ A and MB take the initiative to set up and use an IPSec tunnel
 - ♦ This is transparent to B



Net/host-to-net IPSec VPNs: Mediation with ESP transport mode

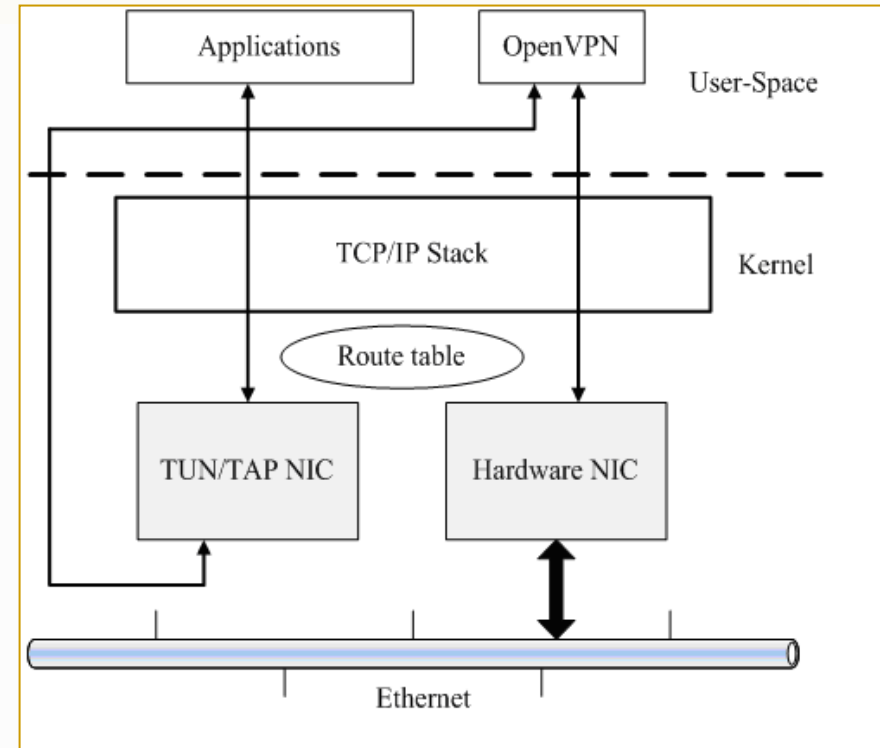


Opportunistic encryption

- ▷ A best-effort approach with legacy support
 - ♦ If a client supports, it uses a secure communication protocol
 - ♦ Otherwise, there is a fallback, insecure alternative
 - ♦ Sometimes only protect against passive wiretapping
- ▷ Usually explored at session/application layer
 - ♦ DNSSEC vs. DNS
 - ♦ HTTPS vs HTTP
 - ♦ STARTTLS for SMTP / IMAP / POP3
- ▷ Some attempts were made for IPSec
 - ♦ But IPSec is a complicated protocol ...

VPN implementation: Special networking devices (1/2)

- ▷ Linux tun/tap
 - ♦ **tun** is an L3 endpoint (gateway / router)
 - ♦ **tap** is an L2 endpoint (bridge)



- ▷ They redirect traffic to an application
 - ♦ The VPN client/server

from <http://ipseclab.eit.lth.se/tiki-index.php?page=6.+OpenVPN>

VPN implementation:

Special networking devices (2/2)

▷ WireGuard tunnel interfaces

- ♦ The tunnels are implemented by WireGuard net interfaces
 - Inside the kernel
- ♦ Tunnel endpoints are identified with public keys
 - And can freely change their IP addresses